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1 Misc

1.1 Template

```
#include <bits/stdc++.h>
#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>
using namespace std;

// random
mt19937 rng(chrono::steady_clock::now().time_since_epoch().count());

// pb_ds stuff
using namespace __gnu_pbds;
const int RANDOM = chrono::high_resolution_clock::now().time_since_epoch().count();
struct chash { int operator()(int x) const { return x ^ RANDOM; } };

template<class T> using ordered_set = tree<T, null_type, less<T>, rb_tree_tag, tree_order_statistics_node_update>;
template<class K, class V> using gpmap = gp_hash_table<K, V, chash>;

int main(){
    ios::sync_with_stdio(false);
    cin.tie(0);
}
```

1.2 Random Primes

```
313, 100003, 200003, 1000003, 10000019, 15000017,
2147483647, 10000000087, 1878234273849638459,
2878234273849638443, 8878234273849638443,
9196969696969696969, 9206969696969696969
```

1.3 Pragmas

```
#pragma GCC optimize("Ofast")
#pragma GCC optimize("unroll-loops")
#pragma GCC target("sse,sse2,sse3,ssse3,sse4,popcnt,abm,mmx,avx,tune=native")
#pragma GCC target("fpmath=387")
```

1.4 Fast IO

```
int read(){
    char c;
    do {
        c = getchar_unlocked();
    } while(c < 33);
    int res = 0, mul = 1;
    if(c == '-'){
        mul = -1;
        c = getchar_unlocked();
    }
    while('0' <= c && c <= '9'){
        res = res * 10 + c - '0';
        c = getchar_unlocked();
    }
    return res * mul;
}

void write(int x){
    static char wbuf[10];
    if(x < 0){
        putchar_unlocked('-');
        x = -x;
    }
    int idx = 0;
    while(x){
        wbuf[idx++] = x % 10;
        x /= 10;
    }
    if(idx == 0)
        putchar_unlocked('0');
```

```
for(int i = idx-1; i >=0; --i)
    putchar_unlocked(wbuf[i] + '0');
}

void write(constchar* s){
    while(*s){
        putchar_unlocked(*s);
        ++s;
    }
}
```

1.5 Day of Date

```
//0-based
const vector<int> T= {0,3,2,5,0,3,5,1,4,6,2,4};
int day(int d, int m, int y){
    y -= (m < 3);
    return (y + y/4 - y/100 + y/400 + T[m-1] +d) % 7;
}
```

1.6 Number of Days since 1-1-1

```
int rdn(int d, int m, int y){
    if (m < 3) --y, m+=12;
    return 365*y + y/4 - y/100 + y/400 + (153*m - 457)/5 + d - 306;
}
```

1.7 Int to Roman

```
const string R[] = {
    "M", "CM", "D", "CD", "C", "XC", "L",
    "XL", "X", "IX", "V", "IV", "I"
};

const int N[]={
    1000, 900, 500, 400, 100, 90, 50, 40, 10,
    9, 5, 4, 1
}
```

```
};

string to_roman(int x){
    if (x == 0)
        return "0"; //Not decimal 0!
    string res = "";
    for(int i = 0; i < 13; ++i){
        while (x >= N[i])
            x -= N[i];
        res += R[i];
    }
    return res;
}
```

2 Data Structures

2.1 2D Segment Tree

```
struct Segtree2D {
    struct Segtree {
        struct node {
            int l, r, val;
            node* lc, * rc;
            node(int _l, int _r, int _val = INF) : l(_l), r(←
                _r), val(_val),
                lc(NULL), rc(NULL) {}
        };
        typedef node* pnode;

        pnode root;

        Segtree(int l, int r) {
            root = new node(l, r);
        }

        void update(pnode& nw, int x, int val) {
            int l = nw->l, r = nw->r, mid = (l + r) / 2;
            if(l == r)
                nw->val = val;
            else {
                assert(l <= x && x <= r);
                pnode& child = x <= mid ? nw->lc : nw->rc;
                if(!child)
                    child = new node(x, x, val);
                else if(child->l <= x && x <= child->r)
                    update(child, x, val);
                else {
                    do {
                        if(x <= mid)
                            r = mid;
                        else
                            l = mid + 1;
                        mid = (l + r) / 2;
                    } while((x <= mid) == (child->l <= mid));
                    pnode nxt = new node(l, r);
                    if(child->l <= mid)
                        nxt->lc = child;
                    else
                        nxt->rc = child;
                    child = nxt;
                    update(nxt, x, val);
                }
            }
            nw->val = min(nw->lc ? nw->lc->val : INF,
                nw->rc ? nw->rc->val : INF);
        }

        int query(pnode& nw, int x1, int x2) {
            if(!nw)
```

```
        return INF;
        int& l = nw->l, &r = nw->r;
        if(r < x1 || x2 < l)
            return INF;
        if(x1 <= l && r <= x2)
            return nw->val;
        int ret = min(query(nw->lc, x1, x2),
            query(nw->rc, x1, x2));
        return ret;
    }

    void update(int x, int val) {
        assert(root->l <= x && x <= root->r);
        update(root, x, val);
    }

    int query(int l, int r) {
        return query(root, l, r);
    }
};

struct node {
    int l, r;
    Segtree y;
    node* lc, * rc;
    node(int _l, int _r) : l(_l), r(_r), y(0, MAX),
        lc(NULL), rc(NULL) {}
};
typedef node* pnode;

pnode root;

Segtree2D(int l, int r) {
    root = new node(l, r);
}

void update(pnode& nw, int x, int y, int val) {
    int& l = nw->l, &r = nw->r, mid = (l + r) / 2;
    if(l == r)
        nw->y.update(y, val);
    else {
        if(x <= mid) {
            if(!nw->lc)
                nw->lc = new node(l, mid);
            update(nw->lc, x, y, val);
        } else {
            if(!nw->rc)
                nw->rc = new node(mid + 1, r);
            update(nw->rc, x, y, val);
        }
        val = min(nw->lc ? nw->lc->y.query(y, y) : INF,
            nw->rc ? nw->rc->y.query(y, y) : INF);
        nw->y.update(y, val);
    }
}

int query(pnode& nw, int x1, int x2, int y1, int y2) ←
    {
        if(!nw)
            return INF;
        int& l = nw->l, &r = nw->r;
        if(r < x1 || x2 < l)
            return INF;
        if(x1 <= l && r <= x2)
            return nw->y.query(y1, y2);
        int ret = min(query(nw->lc, x1, x2, y1, y2),
            query(nw->rc, x1, x2, y1, y2));
        return ret;
    }

    void update(int x, int y, int val) {
```

```
        assert(root->l <= x && x <= root->r);
        update(root, x, y, val);
    }

    int query(int x1, int x2, int y1, int y2) {
        return query(root, x1, x2, y1, y2);
    }
};
```

2.2 Fenwick RU-RQ

```
void updtRL(int l, int r, ll val) {
    updt(BIT1, l, val), updt(BIT1, r + 1, -val);
    updt(BIT2, l, val * (l - 1)), updt(BIT2, r + 1, -val ←
        * r);
}

ll query(int k) {
    return que(BIT1, k) * k - que(BIT2, k);
}
```

2.3 Persistent Segment Tree

```
class PersistentSegtree{
private:
    int n, ptr, sz;
    struct P {
        int val = 0, l, r;
    };
    vector<P> node;
    vector<int> root;

    int newNode(){
        node.emplace_back();
        return ptr++;
    }

    int copyNode(int idx){
        node.push_back(node[idx]);
        return ptr++;
    }

    int build(int l, int r){
        int idx = newNode();
        if (l == r)
            return idx;
        node[idx].l = build(l, (l+r)/2);
        node[idx].r = build((l+r)/2 + 1, r);
        return idx;
    }

    int update(int idx, int l, int r, int x, int val){
        idx = copyNode(idx);
        if (l == r){
            node[idx].val += val;
            return idx;
        }
        int mid = (l+r)/2;
        if(x <= mid)
            node[idx].l = update(node[idx].l, l, mid, x←
                , val);
        else
            node[idx].r = update(node[idx].r, mid+1, r,←
                x, val);
        node[idx].val = node[node[idx].l].val + ←
            node[node[idx].r].val;
        return idx;
    }

    int query(int idxl, int idxr, int l, int r, int x, ←
        int y){
        if(y < l || r < x) ]
            return 0;
        if(x <= l && r <= y)
```

```

        return node[idxr].val - node[idxl].val;
    int mid = (l+r)/2;
    return query(node[idxl].l, node[idxr].l, l, mid←
        , x, y)
    + query(node[idxl].r, node[idxr].r, mid+1, r, x←
        , y);
}

public:
PersistentSegtree(int _n) : n(_n), ptr(0){
    sz = 30*n;
    node.reserve(sz);
    root.push_back(build(1,n));
}

void update(int x, int val){
    root.push_back(update(root.back(), 1, n, x, val←
        ));
}

int query(int l, int r, int x, int y){
    return query(root[l-1], root[r], 1, n, x, y);
}

};

```

2.4 Segment Tree Beats

```

ll ST1[4*N + 5], ST2[4*N + 5]; //MAX value, SUM Value
ll smax[4*N + 5], occ[4*N + 5]; //2nd Max, occurrence of ←
1st Max
ll arr[N + 5];

void pushup(int idx){
    ST1[idx] = max(ST1[2*idx], ST1[2*idx + 1]);
    ST2[idx] = ST2[2*idx] + ST2[2*idx + 1];
    if (ST1[2*idx] == ST1[2*idx + 1]){
        smax[idx] = max(smax[2*idx], smax[2*idx + 1]);
        occ[idx] = occ[2*idx] + occ[2*idx + 1];
    }
    else{
        smax[idx] = min(ST1[2*idx], ST1[2*idx + 1]);
        smax[idx] = max(smax[idx], max(smax[2*idx],smax←
            [2*idx + 1]));
        if (ST1[2*idx] <= ST1[2*idx + 1])
            occ[idx] = occ[2*idx + 1];
        if (ST1[2*idx] > ST1[2*idx + 1])
            occ[idx]=occ[2*idx];
    }
}

void build(int idx, int l, int r){
    if (l == r){
        ST1[idx] = ST2[idx] = arr[l];
        occ[idx] = 1;
        smax[idx] = -1;
        return;
    }
    int mid = (l+r)>>1;
    build(2*idx, l, mid);
    build(2*idx + 1, mid + 1, r);
    pushup(idx);
}

void pushdown(int idx, ll v){
    if(ST1[idx] <= v) return;
    assert(v > smax[idx]);
    ST2[idx] -= (ll)(ST1[idx] - v) * (ll)occ[idx];
    ST1[idx] = v;
}

void down(int idx, ll v){
    pushdown(2*idx, v);
    pushdown(2*idx + 1, v);
}

```

```

void update(int idx, int l, int r, int le, int ri, ll v←
){
    //update min(Ai,t)
    if (r < le || ri < l || ST1[idx] <= v)
        return;
    if (le <= l && r <= ri && smax[idx] < v){
        pushdown(idx, v);
        return;
    }
    down(idx, ST1[idx]);
    int mid = (l+r)>>1;
    update(2*idx, l, mid, le, ri, v);
    update(2*idx + 1, mid + 1, r, le, ri, v);
    pushup(idx);
}

ll query1(int idx, int l, int r, int le, int ri){
    //sumquery
    if(r < le || ri < l)
        return -1;
    if(le <= l && r <= ri)
        return ST1[idx];
    down(idx, ST1[idx]);
    int mid = (l+r)>>1;
    ll resa = query1(2*idx, l, mid, le, ri);
    ll resb = query1(2*idx + 1, mid + 1, r, le, ri);
    return max(resa, resb);
}

ll query2(int idx, int l, int r, int le, int ri){
    //maxquery
    if(r < le || ri < l)
        return 0;
    if(le <= l && r <= ri)
        return ST2[idx];
    down(idx, ST1[idx]);
    int mid = (l+r)>>1;
    ll resa = query2(2*idx, l, mid, le, ri);
    ll resb = query2(2*idx + 1, mid + 1, r, le, ri);
    return resa + resb;
}

```

2.5 Li-Chao Tree

```

// max li-chao tree
// works for the range [0, MAX - 1]
// if min li-chao tree:
// replace every call to max() with min() and every > ←
with <
// also replace -INF with INF

struct Func {
    ll m, c;
    ll operator()(ll x) {
        return x * m + c;
    }
};

const int MAX = 1e9 + 1;
const ll INF = 1e18;
const Func NIL = {0, -INF};

struct Node {
    Func f;
    Node* lc;
    Node* rc;

    Node() : f(NIL), lc(nullptr), rc(nullptr) {}
    Node(const Node& n) : f(n.f), lc(nullptr), rc(nullptr)←
        ) {}
};

```

```

Node* root = new Node;

void insert(Func f, Node* cur = root, int l = 0, int r ←
    = MAX - 1) {
    int m = l + (r - l) / 2;
    bool left = f(l) > cur->f(l);
    bool mid = f(m) > cur->f(m);
    if(mid)
        swap(f, cur->f);
    if(l != r) {
        if(left != mid) {
            if(!cur->lc)
                cur->lc = new Node(*cur);
            insert(f, cur->lc, l, m);
        }
        else {
            if(!cur->rc)
                cur->rc = new Node(*cur);
            insert(f, cur->rc, m + 1, r);
        }
    }
}

ll query(ll x, Node* cur = root, int l = 0, int r = MAX←
    - 1) {
    if(!cur)
        return -INF;
    if(l == r)
        return cur->f(x);
    int m = l + (r - l) / 2;
    if(x <= m)
        return max(cur->f(x), query(x, cur->lc, l, m));
    else
        return max(cur->f(x), query(x, cur->rc, m + 1, r));
}

```

2.6 Heavy-Light Decomposition

```

// graph 1-based index
class HLD{
private:
    vector<vector<int>> adj;
    vector<int> sz, in, out, nxt, par;
    Segtree DS;
    int timer = -1;

    void dfs (int u, int p){
        sz[u] = 1;
        par[u] = p;
        for (auto &v : adj[u])
            if (v != p){
                dfs(v, u);
                sz[u] += sz[v];
                if (sz[v] >= sz[adj[u][0]])
                    swap(v, adj[u][0]);
            }
    }

    void chain(int u, int p){
        if(p == u)
            nxt[u] = u;
        in[u] = ++timer;
        for (auto &v : adj[u])
            if(v != p){
                nxt[v] = (v == adj[u][0] ? nxt[u] : v);
                chain(v, u);
            }
        out[u] = timer;
    }

public:
    HLD(int n){

```

```
adj.assign(n+1, vector<int>());
sz.assign(n+1, 0);
in.resize(n+1);
out.resize(n+1);
nxt.assign(n+1, 0);
par.assign(n+1, -1);
}
void add_edge(int u, int v){
    adj[u].push_back(v);
    adj[v].push_back(u);
}
void update_path(int u, int v, int x){
    for(; in[nxt[u]] != in[nxt[v]]; v = par[nxt[v←
]]){
        if(in[u] > in[v]) swap(u, v);
        DS.update(in[nxt[v]], in[v], x);
    }
    if(in[u] > in[v]) swap(u, v);
    DS.update(in[u], in[v], x);
}
int query_path(int u, int v){
    int ans = -1;
    for(; in[nxt[u]] != in[nxt[v]]; v = par[nxt[v←
]]){
        if(in[u] > in[v]) swap(u, v);
        ans = max(ans, DS.query(in[nxt[v]], in[v]));
    }
    if(in[u] > in[v]) swap(u, v);
    ans = max(ans, DS.query(in[u], in[v]));
    return ans;
}
void update_subtree(int u, int x){
    DS.update(in[u], out[u], x);
}
int query_subtree(int u){
    return DS.query(in[u], out[u]);
}
void init(){
    dfs(1, 1);
    chain(1, 1);
    DS.init(timer + 1);
}
};
```

2.7 Link-Cut Tree

```
// sz for path queries
// sub, vsub for subtree queries
// x->access() brings x to the top and propagates it,
// x's left subtree is path from x to root
// after access, sub is no. of nodes in CC of x,
// vsub is no. of nodes under x
// use makeRoot() for arbitrary path queries

typedef struct Node* Np;
struct Node {
    Np p, c[2]; // parent, children
    bool flip = 0; // subtree flipped or not
    int vtx;
    ll val, sz; // value in node, # nodes in ←
                current splay tree
    int sub, vsub = 0; // # of nodes in CC
    ll stsum; // sum of all val in the splay ←
                tree
    // after access(): from original root to this node
    Node(int _vtx, int _val = 1) : vtx(_vtx), val(_val) {
        p = c[0] = c[1] = nullptr;
        calc();
    }
};
```

```
friend int getSz(Np x) {
    return x ? x->sz : 0;
}
friend int getSub(Np x) {
    return x ? x->sub : 0;
}
friend ll getStSum(Np x) {
    return x ? x->stsum : 0;
}

void prop() { // lazy prop
    if(!flip)
        return;
    swap(c[0], c[1]);
    flip = 0;
    for(int i = 0; i < 2; i++) {
        if(c[i])
            c[i]->flip ^= 1;
    }
}
void calc() { // recalc vals
    for(int i = 0; i < 2; i++) {
        if(c[i])
            c[i]->prop();
    }
    sz = 1 + getSz(c[0]) + getSZ(c[1]);
    sub = 1 + getSub(c[0]) + getSub(c[1]) + vsub;
    stsum = val + getStSum(c[0]) + getStSum(c[1]);
}

// SPLAY TREE OPERATIONS
int dir() { // p is path-parent pointer
    if(!p)
        return -2;
    for(int i = 0; i < 2; i++) {
        if(p->c[i] == this)
            return i;
    }
    return -1; // -> not in current splay tree
}
// test if root of current splay tree
bool isRoot() {
    return dir() < 0;
}
friend void setLink(Np x, Np y, int d) { // x is orig←
    parent
    if(y)
        y->p = x;
    if(d >= 0)
        x->c[d] = y;
}
void rot() { // assume p and p->p propagated
    assert(!isRoot());
    int x = dir();
    Np pa = p;
    setLink(pa->p, this, pa->dir());
    setLink(pa, c[x ^ 1], x);
    setLink(this, pa, x ^ 1);
    pa->calc();
}
void splay() { // bring this node to the root of ←
    splay tree
    while(!isRoot() && !p->isRoot()) {
        p->p->prop(), p->prop(), prop();
        dir() == p->dir() ? p->rot() : rot();
        rot();
    }
    if(!isRoot())
        p->prop(), prop(), rot();
    prop();
    calc();
}
```

```
}
Np fbo(int b) { // find by order
    prop();
    int z = getSz(c[0]); // of splay tree
    if(b == z) {
        splay();
        return this;
    }
    return b < z ? c[0]->fbo(b) : c[1]->fbo(b - z - 1);
}

// BASE OPERATIONS
// bring this to top of splay tree (not impacting ←
    original tree)
void access() {
    for(Np v = this, pre = nullptr; v; v = v->p) {
        v->splay(); // now switch virtual children
        if(pre)
            v->vsub -= pre->sub;
        if(v->c[1])
            v->vsub += v->c[1]->sub;
        v->c[1] = pre;
        v->calc();
        pre = v;
    }
    splay();
    assert(!c[1]); // right subtree is empty
}
void makeRoot() { // of the original tree
    access();
    flip ^= 1;
    access();
    assert(!c[0] && !c[1]);
}

// QUERIES
friend Np lca(Np x, Np y) {
    if(x == y)
        return x;
    x->access(), y->access();
    if(!x->p)
        return nullptr;
    x->splay();
    return x->p ? : x; // y was below x in latter case
    // access at y did not affect x←
    -> not connected
}
friend bool connected(Np x, Np y) {
    return lca(x, y);
}

// no. of nodes above; distance to root in original ←
    tree
int distRoot() {
    access();
    return getSz(c[0]);
}
Np getRoot() { // get root of LCT component in ←
    original tree
    access();
    Np a = this;
    while(a->c[0])
        a = a->c[0], a->prop();
    a->access();
    return a;
}
Np getPar(int b) { // get b-th parent on path to root
    access();
    b = getSz(c[0]) - b;
    assert(b >= 0);
    return fbo(b);
}
// can also get min, max on path to root, etc
```

```
// MODIFICATIONS
void setVal(int v) {
    access();
    val = v;
    calc();
}
void addVal(int v) {
    access();
    val += v;
    calc();
}
}
friend void link(Np x, Np y, bool force = 1) {
    assert(!connected(x, y));
    if(force) {
        y->makeRoot(); /// make x par of y; x -> y
    } else {
        y->access();
        assert(!y->c[0]);
    }
    x->access();
    setLink(y, x, 0);
    y->calc();
}
friend void cut(Np y) { // cut y from its parent
    y->access();
    assert(y->c[0]);
    y->c[0]->p = NULL;
    y->c[0] = NULL;
    y->calc();
}
friend void cut(Np x, Np y) { // if x, y adj in tree
    x->makeRoot();
    y->access();
    assert(y->c[0] == x && !x->c[0] && !x->c[1]);
    cut(y);
}
};
Np LCT[maxV];
```

2.8 Treap

```
// Complexity: O(log N) for split and merge
//
// empty treap: Treap* tr = nullptr;
// insert v at x: [l, r] = split(tr, x), m = Treap(v), ←
//             merge lmr
// delete at x: [l, r] = split(tr, x), [m, r] = split(r←
//             , 1), merge lr
// lazy prop: propagate every time a node is accessed

mt19937_64 rng(chrono::steady_clock::now().←
    time_since_epoch().count());

using Key = int;

struct Treap {
    Key val;
    Treap* left;
    Treap* right;
    int prio, sz;
    Treap() {}
    Treap(int _val);
};

int size(Treap* tr) {
    return tr ? tr->sz : 0;
}

void update(Treap* tr) {
```

```
    tr->sz = 1 + size(tr->left) + size(tr->right);
}

Treap::Treap(Key _val) :
    val(_val), left(nullptr), right(nullptr), prio(rng())←
    {
        update(this);
    }

pair<Treap*, Treap*> split(Treap* tr, int sz) {
    if(!tr) return {nullptr, nullptr};
    int left_sz = size(tr->left);
    if(sz <= left_sz) {
        auto [left, mid] = split(tr->left, sz);
        tr->left = mid;
        update(tr);
        return {left, tr};
    } else {
        auto [mid, right] = split(tr->right, sz - left_sz ←
            1);
        tr->right = mid;
        update(tr);
        return {tr, right};
    }
}

Treap* merge(Treap* l, Treap* r) {
    if(!l)
        return r;
    if(!r)
        return l;
    if(l->prio < r->prio) {
        l->right = merge(l->right, r);
        update(l);
        return l;
    } else {
        r->left = merge(l, r->left);
        update(r);
        return r;
    }
}
```

2.9 Unordered Map Custom Hash

```
struct custom_hash {
    static uint64_t splitmix64(uint64_t x) {
        x += 0x9e3779b97f4a7c15;
        x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
        x = (x ^ (x >> 27)) * 0x94d049bb133111eb;
        return x ^ (x >> 31);
    }
    size_t operator()(uint64_t x) const {
        static const uint64_t FIXED_RANDOM =
            chrono::steady_clock::now().time_since_epoch().←
            count();
        return splitmix64(x + FIXED_RANDOM);
    }
};

unordered_map<int, int, custom_hash> umap;
```

2.10 Mo's on Tree

$ST(u) \leq ST(v)$
 $P = LCA(u, v)$
 If $P = u$, query $[ST(u), ST(v)]$
 Else query $[EN(u), ST(v)] + [ST(P), ST(P)]$

3 Dynamic Programming

3.1 DP Convex Hull

```
/* dp[i] = min k<i {dp[k] + x[i]*m[k]}
   Make sure gradient (m[i]) is either non-increasing ←
   if min,
   or non-decreasing if max. x[i] must be non-←
   decreasing. just sort */
int y[N], m[N];
// while this is true, pop back from dq. a=new line, b=←
// last, c=2nd last
bool cekx(int a, int b, int c) {
    // if not enough, change to cross mul
    // if cross mul, beware of negative denominator, and ←
    // overflow
    return (double)(y[b] - y[a]) / (m[a] - m[b]) <= (←
        double)(y[c] - y[b]) /
        (m[b] - m[c]);
}
```

3.2 DP DNC

```
void f(int rem, int l, int r, int optl, int optr) {
    if(l > r)
        return;
    int mid = l + r >> 1;
    int opt = MOD, optid = mid;
    for(int i = optl; i <= mid && i <= optr; ++i) {
        if(dp[rem - 1][i] + c[i][mid] < opt) {
            opt = dp[rem - 1][i] + c[i][mid];
            optid = i;
        }
    }
    dp[rem][mid] = opt;
    f(rem, l, mid - 1, optl, optid);
    f(rem, mid + 1, r, optid, optr);
    return;
}
rep(i, 1, n)dp[1][i] = c[0][i];
rep(i, 2, k)f(i, i, n, i, n);
```

3.3 DP Knuth-Yao

```
// opt[i+1][j] <= opt[i][j] <= opt[i][j+1]
// dp[i][j] = min{k} dp[i][k]+dp[k][j]+cost[i][j]
for(int k = 0; k <= n; k++) {
    for(int i = 0; i + k <= n; i++) {
        if(k < 2)
            dp[i][i + k] = 0, opt[i][i + k] = i;
        else {
            int sta = opt[i][i + k - 1];
            int end = opt[i + 1][i + k];
            for(int j = sta; j <= end; j++) {
                if(dp[i][j] + dp[j][i + k] + cost[i][i + k] < ←
                    dp[i][i + k]) {
                    dp[i][i + k] = dp[i][j] + dp[j][i + k] + cost←
                        [i][i + k];
                    opt[i][i + k] = j;
                }
            }
        }
    }
}
```

4 Geometry

4.1 Geometry Template

```

/*
TABLE OF CONTENT
0. Basic Rule
    0.1. Everything is in double
    0.2. Every comparison use EPS
    0.3. Every degree in rad
1. General Double Operation
    1.1. const double EPS=1E-9
    1.2. const double PI=acos(-1.0)
    1.3. const double INFD=1E9
    1.3. between_d(double x,double l,double r)
        check whether x is between l and r inclusive ←
        with EPS
    1.4. same_d(double x,double y)
        check whether x=y with EPS
    1.5. dabs(double x)
        absolute value of x
2. Point
    2.1. struct point
        2.1.1. double x,y
            cartesian coordinate of the point
        2.1.2. point()
            default constructor
        2.1.3. point(double _x,double _y)
            constructor, set the point to (_x,_y)
        2.1.4. bool operator< (point other)
            regular pair<double,double> operator < with←
            EPS
        2.1.5. bool operator== (point other)
            regular pair<double,double> operator == ←
            with EPS
    2.2. hypot(point P)
        length of hypotenuse of point P to (0,0)
    2.3. e_dist(point P1,point P2)
        euclidean distance from P1 to P2
    2.4. m_dist(point P1,point P2)
        manhattan distance from P1 to P2
    2.5. point rotate(point P,point O,double angle)
        rotate point P from the origin O by angle ccw
3. Vector
    3.1. struct vec
        3.1.1. double x,y
            x and y magnitude of the vector
        3.1.2. vec()
            default constructor
        3.1.3. vec(double _x,double _y)
            constructor, set the vector to (_x,_y)
        3.1.4. vec(point A,point B)
            constructor, set the vector to vector AB (A←
            →B)
*/
/*General Double Operation*/

const double PI = acos(-1.0);
const double INFD = 1E9;
double between_d(double x, double l, double r) {
    return (min(l, r) <= x + EPS && x <= max(l, r) + EPS)←
    ;
}
double same_d(double x, double y) {
    return between_d(x, y, y);
}
double dabs(double x) {
    if(x < EPS)
        return -x;
    return x;
}

```

```

}
/*Point*/
struct point {
    double x, y;
    point() {
        x = y = 0.0;
    }
    point(double _x, double _y) {
        x = _x;
        y = _y;
    }
    bool operator< (point other) {
        if(x < other.x + EPS)
            return true;
        if(x + EPS > other.x)
            return false;
        return y < other.y + EPS;
    }
    bool operator== (point other) {
        return same_d(x, other.x) && same_d(y, other.y);
    }
};
double e_dist(point P1, point P2) {
    return hypot(P1.x - P2.x, P1.y - P2.y);
}
double m_dist(point P1, point P2) {
    return dabs(P1.x - P2.x) + dabs(P1.y - P2.y);
}
double pointBetween(point P, point L, point R) {
    return (e_dist(L, P) + e_dist(P, R) == e_dist(L, R));
}
bool collinear(point P, point L,
               point R) { //newly added(luis), cek 3 ←
                           poin segaris
    return P.x * (L.y - R.y) + L.x * (R.y - P.y) + R.x * ←
    (P.y - L.y) ==
    0; // bole gnti "dabs(x)<"EPS
}

/*Vector*/
struct vec {
    double x, y;
    vec() {
        x = y = 0.0;
    }
    vec(double _x, double _y) {
        x = _x;
        y = _y;
    }
    vec(point A) {
        x = A.x;
        y = A.y;
    }
    vec(point A, point B) {
        x = B.x - A.x;
        y = B.y - A.y;
    }
};
vec scale(vec v, double s) {
    return vec(v.x * s, v.y * s);
}
vec flip(vec v) {
    return vec(-v.x, -v.y);
}
double dot(vec u, vec v) {
    return (u.x * v.x + u.y * v.y);
}
double cross(vec u, vec v) {
    return (u.x * v.y - u.y * v.x);
}
double norm_sq(vec v) {

```

```

    return (v.x * v.x + v.y * v.y);
}
point translate(point P, vec v) {
    return point(P.x + v.x, P.y + v.y);
}
point rotate(point P, point O, double angle) {
    vec v(0);
    P = translate(P, flip(v));
    return translate(point(P.x * cos(angle) - P.y * sin(←
    angle),
                        P.x * sin(angle) + P.y * cos(←
    angle)), v);
}
point mid(point P, point Q) {
    return point((P.x + Q.x) / 2, (P.y + Q.y) / 2);
}
double angle(point A, point O, point B) {
    vec OA(O, A), OB(O, B);
    return acos(dot(OA, OB) / sqrt(norm_sq(OA) * norm_sq(←
    OB)));
}
int orientation(point P, point Q, point R) {
    vec PQ(P, Q), PR(P, R);
    double c = cross(PQ, PR);
    if(c < -EPS)
        return -1;
    if(c > EPS)
        return 1;
    return 0;
}
/*Line*/
struct line {
    double a, b, c;
    line() {
        a = b = c = 0.0;
    }
    line(double _a, double _b, double _c) {
        a = _a;
        b = _b;
        c = _c;
    }
    line(point P1, point P2) {
        if(P1 < P2)
            swap(P1, P2);
        if(same_d(P1.x, P2.x))
            a = 1.0, b = 0.0, c = -P1.x;
        else
            a = -(P1.y - P2.y) / (P1.x - P2.x), b = 1.0, c = ←
            -(a * P1.x) - P1.y;
    }
    line(point P, double slope) {
        if(same_d(slope, INFD))
            a = 1.0, b = 0.0, c = -P.x;
        else
            a = -slope, b = 1.0, c = -(a * P.x) - P.y;
    }
    bool operator== (line other) {
        return same_d(a, other.a) && same_d(b, other.b) && ←
        same_d(c, other.c);
    }
    double slope() {
        if(same_d(b, 0.0))
            return INFD;
        return -(a / b);
    }
};
bool paralel(line L1, line L2) {
    return same_d(L1.a, L2.a) && same_d(L1.b, L2.b);
}
bool intersection(line L1, line L2, point& P) {
    if(paralel(L1, L2))

```



```

    return false;
    P.x = (L2.b * L1.c - L1.b * L2.c) / (L2.a * L1.b - L1.a * L2.b);
    if(same_d(L1.b, 0.0))
        P.y = -(L2.a * P.x + L2.c);
    else
        P.y = -(L1.a * P.x + L1.c);
    return true;
}
double pointToLine(point P, point A, point B, point& C){
    {
        vec AP(A, P), AB(A, B);
        double u = dot(AP, AB) / norm_sq(AB);
        C = translate(A, scale(AB, u));
        return e_dist(P, C);
    }
}
double lineToLine(line L1, line L2) {
    if(!paralel(L1, L2))
        return 0.0;
    return dabs(L2.c - L1.c) / sqrt(L1.a * L1.a + L1.b * L1.b);
}
/*Line Segment*/
struct segment {
    point P, Q;
    line L;
    segment() {
        point T1;
        P = Q = T1;
        line T2;
        L = T2;
    }
    segment(point _P, point _Q) {
        P = _P;
        Q = _Q;
        if(Q < P)
            swap(P, Q);
        line T(P, Q);
        L = T;
    }
    bool operator==(segment other) {
        return P == other.P && Q == other.Q;
    }
};
bool onSegment(point P, segment S) {
    if(orientation(S.P, S.Q, P) != 0)
        return false;
    return between_d(P.x, S.P.x, S.Q.x) && between_d(P.y, S.P.y, S.Q.y);
}
bool s_intersection(segment S1, segment S2) {
    double o1 = orientation(S1.P, S1.Q, S2.P);
    double o2 = orientation(S1.P, S1.Q, S2.Q);
    double o3 = orientation(S2.P, S2.Q, S1.P);
    double o4 = orientation(S2.P, S2.Q, S1.Q);
    if(o1 != o2 && o3 != o4)
        return true;
    if(o1 == 0 && onSegment(S2.P, S1))
        return true;
    if(o2 == 0 && onSegment(S2.Q, S1))
        return true;
    if(o3 == 0 && onSegment(S1.P, S2))
        return true;
    if(o4 == 0 && onSegment(S1.Q, S2))
        return true;
    return false;
}
double pointToSegment(point P, point A, point B, point& C){
    {
        vec AP(A, P), AB(A, B);
        double u = dot(AP, AB) / norm_sq(AB);

```

```

    if(u < EPS) {
        C = A;
        return e_dist(P, A);
    }
    if(u + EPS > 1.0) {
        C = B;
        return e_dist(P, B);
    }
    return pointToLine(P, A, B, C);
}
double segmentToSegment(segment S1, segment S2) {
    if(s_intersection(S1, S2))
        return 0.0;
    double ret = INF;
    point dummy;
    ret = min(ret, pointToSegment(S1.P, S2.P, S2.Q, dummy));
    ret = min(ret, pointToSegment(S1.Q, S2.P, S2.Q, dummy));
    ret = min(ret, pointToSegment(S2.P, S1.P, S1.Q, dummy));
    ret = min(ret, pointToSegment(S2.Q, S1.P, S1.Q, dummy));
    return ret;
}
/*Circle*/
struct circle {
    point P;
    double r;
    circle() {
        point P1;
        P = P1;
        r = 0.0;
    }
    circle(point _P, double _r) {
        P = _P;
        r = _r;
    }
    circle(point P1, point P2) {
        P = mid(P1, P2);
        r = e_dist(P, P1);
    }
    circle(point P1, point P2, point P3) {
        vector<point> T;
        T.clear();
        T.pb(P1);
        T.pb(P2);
        T.pb(P3);
        sort(T.begin(), T.end());
        P1 = T[0];
        P2 = T[1];
        P3 = T[2];
        point M1, M2;
        M1 = mid(P1, P2);
        M2 = mid(P2, P3);
        point Q2, Q3;
        Q2 = rotate(P2, P1, PI / 2);
        Q3 = rotate(P3, P2, PI / 2);
        vec P1Q2(P1, Q2), P2Q3(P2, Q3);
        point M3, M4;
        M3 = translate(M1, P1Q2);
        M4 = translate(M2, P2Q3);
        line L1(M1, M3), L2(M2, M4);
        intersection(L1, L2, P);
        r = e_dist(P, P1);
    }
    bool operator==(circle other) {
        return (P == other.P && same_d(r, other.r));
    }
};
bool insideCircle(point P, circle C) {

```

```

    return e_dist(P, C.P) <= C.r + EPS;
}
bool c_intersection(circle C1, circle C2, point& P1, point& P2) {
    double d = e_dist(C1.P, C2.P);
    if(d > C1.r + C2.r) {
        return false; //d+EPS kalo butuh
    }
    if(d < dabs(C1.r - C2.r) + EPS)
        return false;
    double x1 = C1.P.x, y1 = C1.P.y, r1 = C1.r, x2 = C2.P.x, y2 = C2.P.y, r2 = C2.r;
    double a = (r1 * r1 - r2 * r2 + d * d) / (2 * d), h = sqrt(r1 * r1 - a * a);
    point T(x1 + a * (x2 - x1) / d, y1 + a * (y2 - y1) / d);
    P1 = point(T.x - h * (y2 - y1) / d, T.y + h * (x2 - x1) / d);
    P2 = point(T.x + h * (y2 - y1) / d, T.y - h * (x2 - x1) / d);
    return true;
}
bool lc_intersection(line L, circle O, point& P1, point& P2) {
    double a = L.a, b = L.b, c = L.c, x = O.P.x, y = O.P.y, r = O.r;
    double A = a * a + b * b, B = 2 * a * b * y - 2 * a * c - 2 * b * b * x,
        C = b * b * x * x + b * b * y * y - 2 * b * c * y + c * c - b * b * r * r;
    double D = B * B - 4 * A * C;
    point T1, T2;
    if(same_d(b, 0.0)) {
        T1.x = c / a;
        if(dabs(x - T1.x) + EPS > r)
            return false;
        if(same_d(T1.x - r - x, 0.0) || same_d(T1.x + r - x, 0.0)) {
            P1 = P2 = point(T1.x, y);
            return true;
        }
        double dx = dabs(T1.x - x), dy = sqrt(r * r - dx * dx);
        P1 = point(T1.x, y - dy);
        P2 = point(T1.x, y + dy);
        return true;
    }
    if(same_d(D, 0.0)) {
        T1.x = -B / (2 * A);
        T1.y = (c - a * T1.x) / b;
        P1 = P2 = T1;
        return true;
    }
    if(D < EPS)
        return false;
    D = sqrt(D);
    T1.x = (-B - D) / (2 * A);
    T1.y = (c - a * T1.x) / b;
    P1 = T1;
    T2.x = (-B + D) / (2 * A);
    T2.y = (c - a * T2.x) / b;
    P2 = T2;
    return true;
}
bool sc_intersection(segment S, circle C, point& P1, point& P2) {
    bool cek = lc_intersection(S.L, C, P1, P2);
    if(!cek)
        return false;
    double x1 = S.P.x, y1 = S.P.y, x2 = S.Q.x, y2 = S.Q.y;
}

```

```

bool b1 = between_d(P1.x, x1, x2) && between_d(P1.y, y1, y2);
bool b2 = between_d(P2.x, x1, x2) && between_d(P2.y, y1, y2);
if(P1 == P2)
    return b1;
if(b1 || b2) {
    if(!b1)
        P1 = P2;
    if(!b2)
        P2 = P1;
    return true;
}
return false;
}
/*Triangle*/
double t_perimeter(point A, point B, point C) {
    return e_dist(A, B) + e_dist(B, C) + e_dist(C, A);
}
double t_area(point A, point B, point C) {
    double s = t_perimeter(A, B, C) / 2;
    double ab = e_dist(A, B), bc = e_dist(B, C), ac = e_dist(C, A);
    return sqrt(s * (s - ab) * (s - bc) * (s - ac));
}
circle t_inCircle(point A, point B, point C) {
    vector<point> T;
    T.clear();
    T.pb(A);
    T.pb(B);
    T.pb(C);
    sort(T.begin(), T.end());
    A = T[0];
    B = T[1];
    C = T[2];
    double r = t_area(A, B, C) / (t_perimeter(A, B, C) / 2);
    double ratio = e_dist(A, B) / e_dist(A, C);
    vec BC(B, C);
    BC = scale(BC, ratio / (1 + ratio));
    point P;
    P = translate(B, BC);
    line AP1(A, P);
    ratio = e_dist(B, A) / e_dist(B, C);
    vec AC(A, C);
    AC = scale(AC, ratio / (1 + ratio));
    P = translate(A, AC);
    line BP2(B, P);
    intersection(AP1, BP2, P);
    return circle(P, r);
}
circle t_outCircle(point A, point B, point C) {
    return circle(A, B, C);
}
/*Polygon*/
struct polygon {
    vector<point> P;
    polygon() {
        P.clear();
    }
    polygon(vector<point>& _P) {
        P = _P;
    }
};
bool rayCast(point P, polygon& A) {
    point Q(P.x, 10000);
    line cast(P, Q);
    int cnt = 0;
    FOR(i, (int)(A.P.size()) - 1) {
        line temp(A.P[i], A.P[i + 1]);
        point I;

```

```

bool B = intersection(cast, temp, I);
if(!B)
    continue;
else if(I == A.P[i] || I == A.P[i + 1])
    continue;
else if(pointBetween(I, A.P[i], A.P[i + 1]) && pointBetween(I, P, Q))
    cnt++;
}
return cnt % 2 == 1;
}
// line segment p-q intersect with line A-B.
point lineIntersectSeg(point p, point q, point A, point B) {
    double a = B.y - A.y;
    double b = A.x - B.x;
    double c = B.x * A.y - A.x * B.y;
    double u = fabs(a * p.x + b * p.y + c);
    double v = fabs(a * q.x + b * q.y + c);
    return point((p.x * v + q.x * u) / (u + v), (p.y * v + q.y * u) / (u + v));
}
// cuts polygon Q along the line formed by point a -> point b
// (note: the last point must be the same as the first point)
vector<point> cutPolygon(point a, point b, const vector<point>& Q) {
    vector<point> P;
    for(int i = 0; i < (int)Q.size(); i++) {
        double left1 = cross(toVec(a, b), toVec(a, Q[i]));
        double left2 = 0;
        if(i != (int)Q.size() - 1)
            left2 = cross(toVec(a, b), toVec(a, Q[i + 1]));
        if(left1 > -EPS)
            P.push_back(Q[i]);
        if(left1 * left2 < -EPS)
            P.push_back(lineIntersectSeg(Q[i], Q[i + 1], a, b));
    }
    if(!P.empty() && !(P.back() == P.front()))
        P.push_back(P.front());
    return P;
}
circle minCoverCircle(polygon& A) {
    vector<point> p = A.P;
    point c;
    circle ret;
    double cr = 0.0;
    int i, j, k;
    c = p[0];
    for(i = 1; i < p.size(); i++) {
        if(e_dist(p[i], c) >= cr + EPS) {
            c = p[i], cr = 0;
            ret = circle(c, cr);
            for(j = 0; j < i; j++) {
                if(e_dist(p[j], c) >= cr + EPS) {
                    c = mid(p[i], p[j]);
                    cr = e_dist(p[i], c);
                    ret = circle(c, cr);
                    for(k = 0; k < j; k++) {
                        if(e_dist(p[k], c) >= cr + EPS) {
                            ret = circle(p[i], p[j], p[k]);
                            c = ret.P;
                            cr = ret.r;
                        }
                    }
                }
            }
        }
    }
}

```

```

return ret;
}
/*Geometry Algorithm*/
double DP[110][110];
double minCostPolygonTriangulation(polygon& A) {
    if(A.P.size() < 3)
        return 0;
    FOR(i, A.P.size()) {
        for(int j = 0, k = i; k < A.P.size(); j++, k++) {
            if(k < j + 2)
                DP[j][k] = 0.0;
            else {
                DP[j][k] = INF;
                REP(l, j + 1, k - 1) {
                    double cost = e_dist(A.P[j], A.P[k]) + e_dist(A.P[k], A.P[l]) + e_dist(A.P[l], A.P[j]);
                    DP[j][k] = min(DP[j][k], DP[j][l] + DP[l][k] + cost);
                }
            }
        }
    }
    return DP[0][A.P.size() - 1];
}

```

4.2 Convex Hull

```

typedef double TD; // for precision
namespace GEOM {
    typedef pair<TD, TD> Pt; // vector and points
    const TD EPS = 1e-9;
    const TD maxD = 1e9;
    TD cross(Pt a, Pt b, Pt c) { // right hand rule
        TD v1 = a.first - c.first; // (a-c) X (b-c)
        TD v2 = a.second - c.second;
        TD u1 = b.first - c.first;
        TD u2 = b.second - c.second;
        return v1 * u2 - v2 * u1;
    }
    TD cross(Pt a, Pt b) { // a X b
        return a.first * b.second - a.second * b.first;
    }
    TD dot(Pt a, Pt b, Pt c) { // (a-c) . (b-c)
        TD v1 = a.first - c.first;
        TD v2 = a.second - c.second;
        TD u1 = b.first - c.first;
        TD u2 = b.second - c.second;
        return v1 * u1 + v2 * u2;
    }
    TD dot(Pt a, Pt b) { // a . b
        return a.first * b.first + a.second * b.second;
    }
    TD dist(Pt a, Pt b) {
        return sqrt((a.first - b.first) * (a.first - b.first) + (a.second - b.second) * (a.second - b.second));
    }
    TD shoelaceX2(vector<Pt>& convHull) {
        TD ret = 0;
        for(int i = 0, n = convHull.size(); i < n; i++)
            ret += cross(convHull[i], convHull[(i + 1) % n]);
        return ret;
    }
    vector<Pt> createConvexHull(vector<Pt>& points) {
        sort(points.begin(), points.end());
        vector<Pt> ret;
        for(int i = 0; i < points.size(); i++) {

```



```

while(ret.size() > 1 &&
    cross(points[i], ret[ret.size() - 1], ret[←
        ret.size() - 2]) < -EPS)
    ret.pop_back();
ret.push_back(points[i]);
}
for(int i = points.size() - 2, sz = ret.size(); i ←
    >= 0; i--) {
    while(ret.size() > sz &&
        cross(points[i], ret[ret.size() - 1], ret[←
            ret.size() - 2]) < -EPS)
        ret.pop_back();
    if(i == 0)
        break;
    ret.push_back(points[i]);
}
return ret;
bool isInside(Pt pv, vector<Pt>& x) { //using ←
    winding number
    int n = x.size(), wn = 0;
    x.push_back(x[0]);
    for(int i = 0; i < n; ++i) {
        if(((x[i + 1].first <= pv.first && x[i].first ←
            >= pv.first) ||
            (x[i + 1].first >= pv.first && x[i].first ←
                <= pv.first)) &&
            ((x[i + 1].second <= pv.second && x[i].←
                second >= pv.second) ||
            (x[i + 1].second >= pv.second && x[i].←
                second <= pv.second))) {
            if(cross(x[i], x[i + 1], pv) == 0) {
                x.pop_back();
                return true;
            }
        }
    }
}
for(int i = 0; i < n; ++i) {
    if(x[i].second <= pv.second) {
        if(x[i + 1].second > pv.second && cross(x[i],←
            x[i + 1], pv) > 0)
            ++wn;
    } else if(x[i + 1].second <= pv.second && cross←
        (x[i], x[i + 1], pv) < 0)
        --wn;
}
x.pop_back();
return wn != 0;
}
bool isInside(Pt pv, vector<Pt>& x) { //using winding←
    number
    int n = x.size(), wn = 0;
    x.push_back(x[0]);
    for(int i = 0; i < n; ++i) {
        if(((x[i + 1].first <= pv.first && x[i].first >= ←
            pv.first) ||
            (x[i + 1].first >= pv.first && x[i].first <= ←
                pv.first)) &&
            ((x[i + 1].second <= pv.second && x[i].second←
                >= pv.second) ||
            (x[i + 1].second >= pv.second && x[i].second←
                <= pv.second))) {
            if(cross(x[i], x[i + 1], pv) == 0) {
                x.pop_back();
                return true;
            }
        }
    }
}
for(int i = 0; i < n; ++i) {
    if(x[i].second <= pv.second) {

```

```

        if(x[i + 1].second > pv.second && cross(x[i], x←
            [i + 1], pv) > 0)
            ++wn;
    } else if(x[i + 1].second <= pv.second && cross(x←
        [i], x[i + 1], pv) < 0)
        --wn;
}
x.pop_back();
return wn != 0;
}
}

```

4.3 Closest Pair of Points

```

#define fi first
#define se second
typedef pair<int, int> pii;
struct Point {
    int x, y, id;
};
int compareX(const void* a, const void* b) {
    Point* p1 = (Point*)a, * p2 = (Point*)b;
    return (p1->x - p2->x);
}
int compareY(const void* a, const void* b) {
    Point* p1 = (Point*)a, * p2 = (Point*)b;
    return (p1->y - p2->y);
}
double dist(Point p1, Point p2) {
    return sqrt((double)(p1.x - p2.x) * (p1.x - p2.x) +
        (double)(p1.y - p2.y) * (p1.y - p2.y)
    );
}
pair<pii, double> bruteForce(Point P[], int n) {
    double min = 1e8;
    pii ret = pii(-1, -1);
    for(int i = 0; i < n; ++i)
        for(int j = i + 1; j < n; ++j)
            if(dist(P[i], P[j]) < min) {
                ret = pii(P[i].id, P[j].id);
                min = dist(P[i], P[j]);
            }
    return pair<pii, double> (ret, min);
}
pair<pii, double> getmin(pair<pii, double> x, pair<pii,←
    double> y) {
    if(x.fi.fi == -1 && x.fi.se == -1)
        return y;
    if(y.fi.fi == -1 && y.fi.se == -1)
        return x;
    return (x.se < y.se) ? x : y;
}
pair<pii, double> stripClosest(Point strip[], int size,←
    double d) {
    double min = d;
    pii ret = pii(-1, -1);
    qsort(strip, size, sizeof(Point), compareY);
    for(int i = 0; i < size; ++i)
        for(int j = i + 1; j < size && (strip[j].y - strip[←
            i].y) < min; ++j)
            if(dist(strip[i], strip[j]) < min) {
                ret = pii(strip[i].id, strip[j].id);
                min = dist(strip[i], strip[j]);
            }
    return pair<pii, double>(ret, min);
}
pair<pii, double> closestUtil(Point P[], int n) {
    if(n <= 3)
        return bruteForce(P, n);
    int mid = n / 2;

```

```

    Point midPoint = P[mid];
    pair<pii, double> dl = closestUtil(P, mid);
    pair<pii, double> dr = closestUtil(P + mid, n - mid);
    pair<pii, double> d = getmin(dl, dr);
    Point strip[n];
    int j = 0;
    for(int i = 0; i < n; i++)
        if(abs(P[i].x - midPoint.x) < d.second)
            strip[j] = P[i], j++;
    return getmin(d, stripClosest(strip, j, d.second));
}
pair<pii, double> closest(Point P[], int n) {
    qsort(P, n, sizeof(Point), compareX);
    return closestUtil(P, n);
}
Point P[50005];
int main() {
    int n;
    scanf("%d", &n);
    for(int a = 0; a < n; a++) {
        scanf("%d%d", &P[a].x, &P[a].y);
        P[a].id = a;
    }
    pair<pii, double> hasil = closest(P, n);
    if(hasil.fi.fi > hasil.fi.se)
        swap(hasil.fi.fi, hasil.fi.se);
    printf("%d %d %.6lf\n", hasil.fi.fi, hasil.fi.se, ←
        hasil.se);
    return 0;
}

```

4.4 Smallest Enclosing Circle

```

// welzl's algo to find the 2d minimum enclosing circle←
// of a set of points
// expected O(N)
// directions: remove duplicates and shuffle points, ←
// then call welzl(points)

struct Point {
    double x;
    double y;
};

struct Circle {
    double x, y, r;
    Circle() {}
    Circle(double _x, double _y, double _r): x(_x), y(_y)←
        , r(_r) {}
};

Circle trivial(const vector<Point>& r) {
    if(r.size() == 0)
        return Circle(0, 0, -1);
    else if(r.size() == 1)
        return Circle(r[0].x, r[0].y, 0);
    else if(r.size() == 2) {
        double cx = (r[0].x + r[1].x) / 2.0, cy = (r[0].y + ←
            r[1].y) / 2.0;
        double rad = hypot(r[0].x - r[1].x, r[0].y - r[1].y)←
            / 2.0;
        return Circle(cx, cy, rad);
    } else {
        double x0 = r[0].x, x1 = r[1].x, x2 = r[2].x;
        double y0 = r[0].y, y1 = r[1].y, y2 = r[2].y;
        double d = (x0 - x2) * (y1 - y2) - (x1 - x2) * (y0 ←
            - y2);
        double cx = (((x0 - x2) * (x0 + x2) + (y0 - y2) * (←
            y0 + y2)) / 2 *

```

```

        (y1 - y2) - ((x1 - x2) * (x1 + x2) + (←
            y1 - y2) * (y1 + y2)) / 2
        * (y0 - y2)) / d;
    double cy = (((x1 - x2) * (x1 + x2) + (y1 - y2) * (←
        y1 + y2)) / 2 *
        (x0 - x2) - ((x0 - x2) * (x0 + x2) + (←
            y0 - y2) * (y0 + y2)) / 2
        * (x1 - x2)) / d;
    return Circle(cx, cy, hypot(x0 - cx, y0 - cy));
}
// SHUFFLE THE POINTS FIRST!!!!!!
Circle welzl(const vector<Point>& p, int idx = 0, ←
    vector<Point> r = {}) {
    if(idx == (int) p.size() || r.size() == 3)
        return trivial(r);
    Circle d = welzl(p, idx + 1, r);
    if(hypot(p[idx].x - d.x, p[idx].y - d.y) > d.r) {
        r.push_back(p[idx]);
        d = welzl(p, idx + 1, r);
    }
    return d;
}

```

4.5 Sutherland-Hodgman Algorithm

```

// Complexity: linear time
// Ada 2 poligon, cari poligon intersectionnya
// poly_point = hasilnya, clipper = pemotongnya
#include<bits/stdc++.h>
using namespace std;

const double EPS = 1e-9;

struct point {
    double x, y;
    point(double _x, double _y): x(_x), y(_y) {}
};
struct vec {
    double x, y;
    vec(double _x, double _y): x(_x), y(_y) {}
};

point pivot(0, 0);
vec toVec(point a, point b) {
    return vec(b.x - a.x, b.y - a.y);
}

double dist(point a, point b) {
    return hypot(a.x - b.x, a.y - b.y);
}

double cross(vec a, vec b) {
    return a.x * b.y - a.y * b.x;
}

bool ccw(point p, point q, point r) {
    return cross(toVec(p, q), toVec(p, r)) > 0;
}

bool collinear(point p, point q, point r) {
    return fabs(cross(toVec(p, q), toVec(p, r))) < EPS;
}

bool lies(point a, point b, point c) {
    if((c.x >= min(a.x, b.x) && c.x <= max(a.x, b.x)) &&
        (c.y >= min(a.y, b.y) && c.y <= max(a.y, b.y)))
        return true;
    else
        return false;
}

bool anglecmp(point a, point b) {
    if(collinear(pivot, a, b))
        return dist(pivot, a) < dist(pivot, b);
}

```

```

double d1x = a.x - pivot.x, d1y = a.y - pivot.y;
double d2x = b.x - pivot.x, d2y = b.y - pivot.y;
return (atan2(d1y, d1x) - atan2(d2y, d2x)) < 0;
}

point intersect(point s1, point e1, point s2, point e2)←
{
    double x1, x2, x3, x4, y1, y2, y3, y4;
    x1 = s1.x;
    y1 = s1.y;
    x2 = e1.x;
    y2 = e1.y;
    x3 = s2.x;
    y3 = s2.y;
    x4 = e2.x;
    y4 = e2.y;
    double num1 = (x1 * y2 - y1 * x2) * (x3 - x4) - (x1 - ←
        x2) * (x3 * y4 - y3 * x4);
    double num2 = (x1 * y2 - y1 * x2) * (y3 - y4) - (y1 - ←
        y2) * (x3 * y4 - y3 * x4);
    double den = (x1 - x2) * (y3 - y4) - (y1 - y2) * (x3 - ←
        x4);
    double new_x = num1 / den;
    double new_y = num2 / den;
    return point(new_x, new_y);
}

void clip(vector<point>& poly_points, point point1, ←
    point point2) {
    vector<point> new_points;
    new_points.clear();
    for(int i = 0; i < poly_points.size(); i++) {
        int k = (i + 1) % poly_points.size();
        double i_pos = ccw(point1, point2, poly_points[i]);
        double k_pos = ccw(point1, point2, poly_points[k]);
        //in in
        if(i_pos <= 0 && k_pos <= 0)
            new_points.push_back(poly_points[k]);
        //out in
        else if(i_pos > 0 && k_pos <= 0) {
            new_points.push_back(intersect(point1, point2, ←
                poly_points[i],
                poly_points[k]));
            new_points.push_back(poly_points[k]);
        }
        // in out
        else if(i_pos <= 0 && k_pos > 0) {
            new_points.push_back(intersect(point1, point2, ←
                poly_points[i],
                poly_points[k]));
        }
        //out out
        else {
        }
    }
    poly_points.clear();
    for(int i = 0; i < new_points.size(); i++)
        poly_points.push_back(new_points[i]);
}

double area(const vector<point>& P) {
    double result = 0.0;
    double x1, y1, x2, y2;
    for(int i = 0; i < P.size() - 1; i++) {
        x1 = P[i].x;
        y1 = P[i].y;
        x2 = P[i + 1].x;
        y2 = P[i + 1].y;
        result += (x1 * y2 - x2 * y1);
    }
    return fabs(result) / 2;
}

```

```

void suthHodgClip(vector<point>& poly_points, vector<←
    point> clipper_points) {
    for(int i = 0; i < clipper_points.size(); i++) {
        int k = (i + 1) % clipper_points.size();
        clip(poly_points, clipper_points[i], clipper_points←
            [k]);
    }
}

vector<point> sortku(vector<point> P) {
    int P0 = 0;
    int i;
    for(i = 1; i < 3; i++) {
        if(P[i].y < P[P0].y || (P[i].y == P[P0].y && P[i].x←
            > P[P0].x))
            P0 = i;
    }
    point temp = P[0];
    P[0] = P[P0];
    P[P0] = temp;
    pivot = P[0];
    sort(++P.begin(), P.end(), anglecmp);
    reverse(++P.begin(), P.end());
    return P;
}

int main {
    clipper_points = sortku(clipper_points);
    suthHodgClip(poly_points, clipper_points);
}

```

4.6 Centroid of Polygon

$$C_x = \frac{1}{6A} \sum_{i=0}^{n-1} (x_i + x_{i+1})(x_i y_{i+1} - x_{i+1} y_i)$$

$$C_y = \frac{1}{6A} \sum_{i=0}^{n-1} (y_i + y_{i+1})(x_i y_{i+1} - x_{i+1} y_i)$$

4.7 Pick Theorem

A: Area of a simply closed lattice polygon

B: Number of lattice points on the edges

I: Number of points in the interior

$$A = I + \frac{B}{2} - 1$$

5 Graphs

5.1 Articulation Point and Bridge

```

// gr -> adj list
// vector vis, low -> initialize to -1
// int timer -> initialize to 0
void dfs(int pos, int dad = -1) {
    vis[pos] = low[pos] = timer++;
    int kids = 0;
    for(auto& i : gr[pos]) {
        if(i == dad)
            continue;
        if(vis[i] >= 0)
            low[pos] = min(low[pos], vis[i]);
        else {
            dfs(i, pos);
            low[pos] = min(low[pos], low[i]);
            if(low[i] > vis[pos])
                is_bridge(pos, i)
                if(low[i] >= vis[pos] && dad >= 0)
                    is_articulation_point(pos)
                    ++kids;
        }
    }
    if(dad == -1 && kids > 1)
}

```

```
is_articulation_point(pos)
}
```

5.2 SCC and Strong Orientation

```
#define N 10020
vector<int> adj[N];
bool vis[N], ins[N];
int disc[N], low[N], gr[N];
stack<int> st;
int id, grid;
void scc(int cur, int par) {
    disc[cur] = low[cur] = ++id;
    vis[cur] = ins[cur] = 1;
    st.push(cur);
    for(int to : adj[cur]) {
        //if (to==par) continue; // ini untuk S0(scc ←
        undirected)
        if(!vis[to])
            scc(to, cur);
        if(ins[to])
            low[cur] = min(low[cur], low[to]);
    }
    if(low[cur] == disc[cur]) {
        grid++; // group id
        while(ins[cur]) {
            gr[st.tp] = grid;
            ins[st.tp] = 0;
            st.pop();
        }
    }
}
```

5.3 Dinic's Maximum Flow

```
// O(VE log(max_flow)) if scaling == 1
// O((V + E) sqrt(E)) if unit graph (turn scaling off)
// O((V + E) sqrt(V)) if bipartite matching (turn ←
    scaling off)
// indices are 0-based
const ll INF = 1e18;

struct Dinic {
    struct Edge {
        int v;
        ll cap, flow;
        Edge(int _v, ll _cap): v(_v), cap(_cap), flow(0) {}
    };

    int n;
    ll lim;
    vector<vector<int>> gr;
    vector<Edge> e;
    vector<int> idx, lv;

    bool has_path(int s, int t) {
        queue<int> q;
        q.push(s);
        lv.assign(n, -1);
        lv[s] = 0;
        while(!q.empty()) {
            int c = q.front();
            q.pop();
            if(c == t)
                break;
            for(auto& i : gr[c]) {
                ll cur_flow = e[i].cap - e[i].flow;
                if(lv[e[i].v] == -1 && cur_flow >= lim) {
                    lv[e[i].v] = lv[c] + 1;

```

```
                q.push(e[i].v);
            }
        }
    }
    return lv[t] != -1;
}

ll get_flow(int s, int t, ll left) {
    if(!left || s == t)
        return left;
    while(idx[s] < (int) gr[s].size()) {
        int i = gr[s][idx[s]];
        if(lv[e[i].v] == lv[s] + 1) {
            ll add = get_flow(e[i].v, t, min(left, e[i].cap ←
                - e[i].flow));
            if(add) {
                e[i].flow += add;
                e[i ^ 1].flow -= add;
                return add;
            }
        }
        ++idx[s];
    }
    return 0;
}

Dinic(int vertices, bool scaling = 1) : // toggle ←
    scaling here
    n(vertices), lim(scaling ? 1 << 30 : 1), gr(n) {}

void add_edge(int from, int to, ll cap, bool directed ←
    = 1) {
    gr[from].push_back(e.size());
    e.emplace_back(to, cap);
    gr[to].push_back(e.size());
    e.emplace_back(from, directed ? 0 : cap);
}

ll get_max_flow(int s, int t) { // call this
    ll res = 0;
    while(lim) { // scaling
        while(has_path(s, t)) {
            idx.assign(n, 0);
            while(ll add = get_flow(s, t, INF))
                res += add;
        }
        lim >>= 1;
    }
    return res;
}
};
```

5.4 Minimum Cost Maximum Flow

```
using FlowT = ll;
using CostT = ll;

const FlowT F_INF = 1e18;
const CostT C_INF = 1e18;
const int MAX_V = 1e5 + 5;
const int MAX_E = 1e6 + 5;

namespace MCMF {
    int n, E;
    int adj[MAX_E], nxt[MAX_E], lst[MAX_V], frm[MAX_V], ←
        vis[MAX_V];
    FlowT cap[MAX_E], flw[MAX_E], totalFlow;
    CostT cst[MAX_E], dst[MAX_V], totalCost;

    void init(int _n) {

```

```
        n = _n;
        fill_n(lst, n, -1), E = 0;
    }
    void add(int u, int v, FlowT ca, CostT co) {
        adj[E] = v, cap[E] = ca, flw[E] = 0, cst[E] = +co;
        nxt[E] = lst[u], lst[u] = E++;
        adj[E] = u, cap[E] = 0, flw[E] = 0, cst[E] = -co;
        nxt[E] = lst[v], lst[v] = E++;
    }
    int spfa(int s, int t) {
        fill_n(dst, n, C_INF), dst[s] = 0;
        queue<int> que;
        que.push(s);
        while(que.size()) {
            int u = que.front();
            que.pop();
            for(int e = lst[u]; e != -1; e = nxt[e])
                if(flw[e] < cap[e]) {
                    int v = adj[e];
                    if(dst[v] > dst[u] + cst[e]) {
                        dst[v] = dst[u] + cst[e];
                        frm[v] = e;
                        if(!vis[v]) {
                            vis[v] = 1;
                            que.push(v);
                        }
                    }
                }
            vis[u] = 0;
        }
        return dst[t] < C_INF;
    }
    pair<FlowT, CostT> solve(int s, int t) {
        totalCost = 0, totalFlow = 0;
        while(1) {
            if(!spfa(s, t))
                break;
            FlowT mn = F_INF;
            for(int v = t, e = frm[v]; v != s; v = adj[e ^ ←
                1], e = frm[v])
                mn = min(mn, cap[e] - flw[e]);
            for(int v = t, e = frm[v]; v != s; v = adj[e ^ ←
                1], e = frm[v]) {
                flw[e] += mn;
                flw[e ^ 1] -= mn;
            }
            totalFlow += mn;
            totalCost += mn * dst[t];
        }
        return {totalFlow, totalCost};
    }
};
```

5.5 Flows with Demands

let S0 be the source and T0 be the original sink

1. add 2 additional nodes, call them S1 and T1
2. connect S0 to nodes normally
3. connect nodes to T0 normally
4. for each edge(U, V), cap = original cap - demand
5. for each node N :
 1. add an edge(S1, N), cap = sum of inward demand to ←
 N
 2. add an edge(N, T1), cap = sum of outward demand ←
 from N
6. add an edge(T0, S0), cap = INF
7. the above is not a typo!
8. run max flow normally
9. for each edge(S1, V) and (U, T1), check if flow == ←
 cap

if step #9 fails, then it is not possible to satisfy $\leftrightarrow$ the given demand

Mathematically, let $d(e)$ be the demand of edge e . Let V be the set of every vertex in the graph.

- $c'(S_1, v) = \sum_{u \in V} d(u, v)$ for each edge (s', v) .
- $c'(v, T_1) = \sum_{v \in V} d(v, w)$ for each edge (v, t') .
- $c'(u, v) = c(u, v) - d(u, v)$ for each edge (u, v) in the old network.
- $c'(T_0, S_0) = \infty$

5.6 Hungarian

```
template <typename TD> struct Hungarian {
    TD INF = 1e9; //max_inf
    int n;
    vector<vector<TD> > adj; // cost[left][right]
    vector<TD> hl, hr, slk;
    vector<int> fl, fr, vl, vr, pre;
    deque<int> q;
    Hungarian(int _n) {
        n = _n;
        adj = vector<vector<TD> >(n, vector<TD>(n, 0));
    }
    int check(int i) {
        if(vl[i] == 1, fl[i] != -1)
            return q.push_back(fl[i]), vr[fl[i]] = 1;
        while(i != -1)
            swap(i, fr[fl[i] = pre[i]]);
        return 0;
    }
    void bfs(int s) {
        slk.assign(n, INF);
        vl.assign(n, 0);
        vr = vl;
        q.assign(vr[s] = 1, s);
        for(TD d; ) {
            for(; !q.empty(); q.pop_front()) {
                for(int i = 0, j = q.front(); i < n; i++) {
                    if(d = hl[i] + hr[j] - adj[i][j], !vl[i] && d <= slk[i]) {
                        if(pre[i] == j, d)
                            slk[i] = d;
                        else if(!check(i))
                            return;
                    }
                }
            }
        }
        d = INF;
        for(int i = 0; i < n; i++) if(!vl[i] && d > slk[i])
            d = slk[i];
        for(int i = 0; i < n; i++) {
            if(vl[i])
                hl[i] += d;
            else
                slk[i] -= d;
            if(vr[i])
                hr[i] -= d;
        }
        for(int i = 0; i < n; i++) if(!vl[i] && !slk[i])
            && !check(i)
            return;
    }
}
TD solve() {
```

```
    fl.assign(n, -1);
    fr = fl;
    hl.assign(n, 0);
    hr = hl;
    pre.assign(n, 0);
    for(int i = 0; i < n; i++)
        hl[i] = *max_element(adj[i].begin(), adj[i].begin()
            () + n);
    for(int i = 0; i < n; i++)
        bfs(i);
    TD ret = 0;
    for(int i = 0; i < n; i++) if(adj[i][fl[i]])
        ret += adj[i][fl[i]];
    return ret;
}
}; //i WILL be matched with fl[i]
```

5.7 Edmonds' Blossom

```
// Maximum matching on general graphs in O(V^2 E)
// Indices are 1-based
// Stolen from ko_osaga's cheatsheet
struct Blossom {
    vector<int> vis, dad, orig, match, aux;
    vector<vector<int>> conn;
    int t, N;
    queue<int> Q;

    void augment(int u, int v) {
        int pv = v;
        do {
            pv = dad[v];
            int nv = match[pv];
            match[v] = pv;
            match[pv] = v;
            v = nv;
        } while(u != pv);
    }

    int lca(int v, int w) {
        ++t;
        while(true) {
            if(v) {
                if(aux[v] == t)
                    return v;
                aux[v] = t;
                v = orig[dad[match[v]]];
            }
            swap(v, w);
        }
    }

    void blossom(int v, int w, int a) {
        while(orig[v] != a) {
            dad[v] = w;
            w = match[v];
            if(vis[w] == 1) {
                Q.push(w);
                vis[w] = 0;
            }
            orig[v] = orig[w] = a;
            v = dad[w];
        }
    }

    bool bfs(int u) {
        fill(vis.begin(), vis.end(), -1);
        iota(orig.begin(), orig.end(), 0);
        Q = queue<int>();
        Q.push(u);
```

```
        vis[u] = 0;
        while(!Q.empty()) {
            int v = Q.front();
            Q.pop();
            for(int x : conn[v]) {
                if(vis[x] == -1) {
                    dad[x] = v;
                    vis[x] = 1;
                    if(!match[x]) {
                        augment(u, x);
                        return 1;
                    }
                    Q.push(match[x]);
                    vis[match[x]] = 0;
                } else if(vis[x] == 0 && orig[v] != orig[x]) {
                    int a = lca(orig[v], orig[x]);
                    blossom(x, v, a);
                    blossom(v, x, a);
                }
            }
        }
        return false;
    }

    Blossom(int n) : // n = vertices
        vis(n + 1), dad(n + 1), orig(n + 1), match(n + 1),
        aux(n + 1), conn(n + 1), t(0), N(n) {
        for(int i = 0; i <= n; ++i) {
            conn[i].clear();
            match[i] = aux[i] = dad[i] = 0;
        }
    }

    void add_edge(int u, int v) {
        conn[u].push_back(v);
        conn[v].push_back(u);
    }

    int solve() { // call this for answer
        int ans = 0;
        vector<int> V(N - 1);
        iota(V.begin(), V.end(), 1);
        shuffle(V.begin(), V.end(), mt19937(0x94949));
        for(auto x : V) {
            if(!match[x]) {
                for(auto y : conn[x]) {
                    if(!match[y]) {
                        match[x] = y, match[y] = x;
                        ++ans;
                        break;
                    }
                }
            }
        }
        for(int i = 1; i <= N; ++i) {
            if(!match[i] && bfs(i))
                ++ans;
        }
        return ans;
    }
};
```

5.8 Eulerian Path or Cycle

```
// finds a eulerian path / cycle
// visits each edge only once
// properties:
// - cycle: degrees are even
// - path: degrees are even OR degrees are even except  $\leftrightarrow$ 
// for 2 vertices
```

```
// how to use: g = adjacency list g[n] = connected to n
, undirected
// if there is a vertex u with an odd degree, call dfs(
u)
// else call on any vertex
// ans = path result

vector<set<int>> g;
vector<int> ans;

void dfs(int u) {
    while(g[u].size()) {
        int v = *g[u].begin();
        g[u].erase(v);
        g[v].erase(u);
        dfs(v);
    }
    ans.push_back(u);
}
```

5.9 Hierholzer's Algorithm

```
// Eulerian on Directed Graph
stack<int> path;
vector<int> euler;
inline void hierholzer() {
    path.push(0);
    int cur = 0;
    while(!path.empty()) {
        if(!adj[cur].empty()) {
            path.push(cur);
            int next = adj[cur].back();
            adj[cur].pop();
            cur = next;
        } else {
            euler.pb(cur);
            cur = path.top();
            path.pop();
        }
    }
    reverse(euler.begin(), euler.end());
}
```

5.10 2-SAT

```
struct TwoSAT {
    int n;
    vector<vector<int>> g, gr;
    vector<int> comp, topological_order, answer;
    vector<bool> vis;

    TwoSAT() {}
    TwoSAT(int _n) :
        n(_n), g(2 * n), gr(2 * n), comp(2 * n), answer(2 *
n), vis(2 * n) {}

    void add_edge(int u, int v) {
        g[u].push_back(v);
        gr[v].push_back(u);
    }

    // For the following three functions
    // int x, bool val: if 'val' is true, we take the
variable to be x.
    // Otherwise we take it to be x's complement.

    // At least one of them is true
    void add_clause_or(int i, bool f, int j, bool p) {
        add_edge(i + (f ? n : 0), j + (p ? 0 : n));
    }
}
```

```
add_edge(j + (p ? n : 0), i + (f ? 0 : n));
}

// Only one of them is true
void add_clause_xor(int i, bool f, int j, bool p) {
    add_clause_or(i, f, j, p);
    add_clause_or(i, !f, j, !p);
}

// Both of them have the same value
void add_clause_and(int i, bool f, int j, bool p) {
    add_clause_xor(i, !f, j, p);
}

// Topological sort
void dfs(int u) {
    vis[u] = true;
    for(const auto& v : g[u])
        if(!vis[v])
            dfs(v);
    topological_order.push_back(u);
}

// Extracting strongly connected components
void scc(int u, int id) {
    vis[u] = true;
    comp[u] = id;
    for(const auto& v : gr[u])
        if(!vis[v])
            scc(v, id);
}

bool satisfiable() {
    fill(vis.begin(), vis.end(), false);
    for(int i = 0; i < 2 * n; i++)
        if(!vis[i])
            dfs(i);
    fill(vis.begin(), vis.end(), false);
    reverse(topological_order.begin(),
topological_order.end());
    int id = 0;
    for(const auto& v : topological_order)
        if(!vis[v])
            scc(v, id++);
    // Constructing the answer
    for(int i = 0; i < n; i++) {
        if(comp[i] == comp[i + n])
            return false;
        answer[i] = (comp[i] > comp[i + n] ? 1 : 0);
    }
    return true;
}
};
```

6 Math

6.1 Extended Euclidean GCD

```
// computes x and y such that ax + by = gcd(a, b) in O(
log(min(a, b)))
// returns {gcd(a, b), x, y}
tuple<int, int, int> gcd(int a, int b) {
    if(b == 0) return {a, 1, 0};
    auto [d, x1, y1] = gcd(b, a % b);
    return {d, y1, x1 - y1 * (a / b)};
}
```

6.2 Generalized CRT

```
template<typename T>
T extended_euclid(T a, T b, T& x, T& y) {
    if(b == 0) {
        x = 1;
        y = 0;
        return a;
    }
    T xx, yy, gcd;
    gcd = extended_euclid(b, a % b, xx, yy);
    x = yy;
    y = xx - (yy * (a / b));
    return gcd;
}

template<typename T>
T MOD(T a, T b) {
    return (a % b + b) % b;
}

// return x, lcm. x = a % n && x = b % m
template<typename T>
pair<T, T> CRT(T a, T n, T b, T m) {
    T _n, _m;
    T gcd = extended_euclid(n, m, _n, _m);
    if(n == m) {
        if(a == b)
            return pair<T, T>(a, n);
        else
            return pair<T, T>(-1, -1);
    } else if(abs(a - b) % gcd != 0)
        return pair<T, T>(-1, -1);
    else {
        T lcm = m * n / gcd;
        T x = MOD(a + MOD(n * MOD(_n * ((b - a) / gcd), m /
gcd), lcm), lcm);
        return pair<T, T>(x, lcm);
    }
}
```

6.3 Generalized Lucas Theorem

```
/*Special Lucas : (n,k) % p^x
fctp[n] = Product of the integers less than or equal
to n that are not divisible by p
Precompute fctp*/
LL p
LL E(LL n, int m) {
    LL tot = 0;
    while(n != 0)
        tot += n / m, n /= m;
    return tot;
}

LL funct(LL n, LL base) {
    LL ans = fast(fctp[base], n / base, base) * fctp[n %
base] % base;
    return ans;
}

LL F(LL n, LL base) {
    LL ans = 1;
    while(n != 0) {
        ans = (ans * funct(n, base)) % base;
        n /= p;
    }
    return ans;
}

LL special_lucas(LL n, LL r, LL base) {
    p = fprime(base);
    LL pow = E(n, p) - E(n - r, p) - E(r, p);
    LL TOP = fast(p, pow, base) * F(n, base) % base;
}
```

```
LL BOT = F(r, base) * F(n - r, base) % base;
return (TOP * fast(BOT, totien(base) - 1, base)) % base;
}
//End of Special Lucas
```

6.4 Linear Diophantine

```
//FOR SOLVING MINIMUM ABS(X) + ABS(Y)
ll x, y, newX, newY, target = 0;
ll extGcd(ll a, ll b) {
    if(b == 0) {
        x = 1, y = 0;
        return a;
    }
    ll ret = extGcd(b, a % b);
    newX = y;
    newY = x - y * (a / b);
    x = newX;
    y = newY;
    return ret;
}
ll fix(ll sol, ll rt) {
    ll ret = 0;
    //CASE SOLUTION(X/Y) < TARGET
    if(sol < target)
        ret = -floor(abs(sol + target) / (double)rt);
    //CASE SOLUTION(X/Y) > TARGET
    if(sol > target)
        ret = ceil(abs(sol - target) / (double)rt);
    return ret;
}
ll work(ll a, ll b, ll c) {
    ll gcd = extGcd(a, b);
    ll solX = x * (c / gcd);
    ll solY = y * (c / gcd);
    a /= gcd;
    b /= gcd;
    ll fi = abs(fix(solX, b));
    ll se = abs(fix(solY, a));
    ll lo = min(fi, se);
    ll hi = max(fi, se);
    ll ans = abs(solX) + abs(solY);
    for(ll i = lo; i <= hi; i++) {
        ans = min(ans, abs(solX + i * b) + abs(solY - i * a));
        ans = min(ans, abs(solX - i * b) + abs(solY + i * a));
    }
    return ans;
}
```

6.5 Modular Linear Equation

```
// finds all solutions to ax = b (mod n)
vi modular_linear_equation_solver(int a, int b, int n) {
    {
        int x, y;
        vi ret;
        int g = extended_euclid(a, n, x, y);
        if(!(b % g)) {
            x = mod(x * (b / g), n);
            for(int i = 0; i < g; i++)
                ret.push_back(mod(x + i * (n / g), n));
        }
        return ret;
    }
}
```

6.6 Miller-Rabin and Pollard's Rho

```
namespace MillerRabin {
    const vector<ll> primes = { // deterministic up to 2^64 - 1
        2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37
    };
    ll gcd(ll a, ll b) {
        return b ? gcd(b, a % b) : a;
    }
    ll powa(ll x, ll y, ll p) { // (x ^ y) % p
        if(!y)
            return 1;
        if(y & 1)
            return ((__int128) x * powa(x, y - 1, p)) % p;
        ll temp = powa(x, y >> 1, p);
        return ((__int128) temp * temp) % p;
    }
    bool miller_rabin(ll n, ll a, ll d, int s) {
        ll x = powa(a, d, n);
        if(x == 1 || x == n - 1)
            return 0;
        for(int i = 0; i < s; ++i) {
            x = ((__int128) x * x) % n;
            if(x == n - 1)
                return 0;
        }
        return 1;
    }
    bool is_prime(ll x) { // use this
        if(x < 2)
            return 0;
        int r = 0;
        ll d = x - 1;
        while((d & 1) == 0) {
            d >>= 1;
            ++r;
        }
        for(auto& i : primes) {
            if(x == i)
                return 1;
            if(miller_rabin(x, i, d, r))
                return 0;
        }
        return 1;
    }
}

namespace PollardRho {
    mt19937_64 generator(chrono::steady_clock::now()
        .time_since_epoch().count());
    uniform_int_distribution<ll> rand_ll(0, LLONG_MAX);
    ll f(ll x, ll b, ll n) { // (x^2 + b) % n
        return (((__int128) x * x) % n + b) % n;
    }
    ll rho(ll n) {
        if(n % 2 == 0)
            return 2;
        ll b = rand_ll(generator);
        ll x = rand_ll(generator);
        ll y = x;
        while(1) {
            x = f(x, b, n);
            y = f(f(y, b, n), b, n);
            ll d = MillerRabin::gcd(abs(x - y), n);
            if(d != 1)
                return d;
        }
    }
    void pollard_rho(ll n, vector<ll>& res) {
        if(n == 1)
```

```
        return;
        if(MillerRabin::is_prime(n)) {
            res.push_back(n);
            return;
        }
        ll d = rho(n);
        pollard_rho(d, res);
        pollard_rho(n / d, res);
    }
    vector<ll> factorize(ll n, bool sorted = 1) { // use this
        vector<ll> res;
        pollard_rho(n, res);
        if(sorted)
            sort(res.begin(), res.end());
        return res;
    }
}
```

6.7 Berlekamp-Massey

```
#include <bits/stdc++.h>
using namespace std;
#define pb push_back
typedef long long ll;
#define SZ 233333
const int MOD = 1e9 + 7; //or any prime
ll qp(ll a, ll b) {
    ll x = 1;
    a %= MOD;
    while(b) {
        if(b & 1)
            x = x * a % MOD;
        a = a * a % MOD;
        b >>= 1;
    }
    return x;
}
namespace linear_seq {
    vector<int> BM(vector<int> x) {
        //ls: (shortest) relation sequence (after filling zeroes) so far
        //cur: current relation sequence
        vector<int> ls, cur;
        //lf: the position of ls (t')
        //ld: delta of ls (v')
        int lf = -1, ld = -1;
        for(int i = 0; i < int(x.size()); ++i) {
            ll t = 0;
            //evaluate at position i
            for(int j = 0; j < int(cur.size()); ++j)
                t = (t + x[i - j - 1] * (ll)cur[j]) % MOD;
            if((t - x[i]) % MOD == 0) {
                continue; //good so far
            }
            //first non-zero position
            if(!cur.size()) {
                cur.resize(i + 1);
                lf = i;
                ld = (t - x[i]) % MOD;
                continue;
            }
            //cur=cur-c/ld*(x[i]-t)
            ll k = -(x[i] - t) * qp(ld, MOD - 2) % MOD /*1/ld*/;
            vector<int> c(i - lf - 1); //add zeroes in front
            c.pb(k);
            for(int j = 0; j < int(ls.size()); ++j)
                c.pb(-ls[j]*k % MOD);
            if(c.size() < cur.size())
```



```

        c.resize(cur.size());
        for(int j = 0; j < int(cur.size()); ++j)
            c[j] = (c[j] + cur[j]) % MOD;
        //if cur is better than ls, change ls to cur
        if(i - lf + (int)ls.size() >= (int)cur.size())
            ls = cur, lf = i, ld = (t - x[i]) % MOD;
        cur = c;
    }
    for(int i = 0; i < int(cur.size()); ++i)
        cur[i] = (cur[i] % MOD + MOD) % MOD;
    return cur;
}
int m; //length of recurrence
//a: first terms
//h: relation
ll a[SZ], h[SZ], t_[SZ], s[SZ], t[SZ];
//calculate p*q mod f
void mull(ll* p, ll* q) {
    for(int i = 0; i < m + m; ++i)
        t_[i] = 0;
    for(int i = 0; i < m; ++i) if(p[i])
        for(int j = 0; j < m; ++j)
            t_[i + j] = (t_[i + j] + p[i] * q[j]) % MOD;
    for(int i = m + m - 1; i >= m; --i) if(t_[i])
        //miuns t_[i]x^{i-m}(x^m-\sum_{j=0}^{m-1} x^{m-1-j}h_j)
        for(int j = m - 1; ~j; --j)
            t_[i - j - 1] = (t_[i - j - 1] + t_[i] * h[j] % MOD) % MOD;
    for(int i = 0; i < m; ++i)
        p[i] = t_[i];
}
ll calc(ll K) {
    for(int i = m; ~i; --i)
        s[i] = t[i] = 0;
    //init
    s[0] = 1;
    if(m != 1)
        t[1] = 1;
    else
        t[0] = h[0];
    //binary-exponentiation
    while(K) {
        if(K & 1)
            mull(s, t);
        mull(t, t);
        K >>= 1;
    }
    ll su = 0;
    for(int i = 0; i < m; ++i)
        su = (su + s[i] * a[i]) % MOD;
    return (su % MOD + MOD) % MOD;
}
int work(vector<int> x, ll n) {
    if(n < int(x.size()))
        return x[n];
    vector<int> v = BM(x);
    m = v.size();
    if(!m)
        return 0;
    for(int i = 0; i < m; ++i)
        h[i] = v[i], a[i] = x[i];
    return calc(n);
}
}
using linear_seq::work;

const vector<int> sequence = {
    0, 2, 2, 28, 60, 836, 2766
};
int main() {

```

```

    cout << work(sequence, 7) << '\n';
}

```

6.8 Fast Fourier Transform

```

using ld = double; // change to long double if reach 10^18
using cd = complex<ld>;
const ld PI = acos(-(ld)1);

void fft(vector<cd>& a, int sign = 1) {
    int n = a.size();
    ld theta = sign * 2 * PI / n;
    for(int i = 0, j = 1; j < n - 1; j++) {
        for(int k = n >> 1; k > (i ^= k); k >>= 1);
        if(j < i)
            swap(a[i], a[j]);
    }
    for(int m, mh = 1; (m = mh << 1) <= n; mh = m) {
        int irev = 0;
        for(int i = 0; i < n; i += m) {
            cd w = exp(cd(0, theta * irev));
            for(int k = n >> 2; k > (irev ^= k); k >>= 1);
            for(int j = i; j < mh + i; j++) {
                int k = j + mh;
                cd x = a[j] - a[k];
                a[j] += a[k];
                a[k] = w * x;
            }
        }
        if(sign == -1) for(cd& i : a)
            i /= n;
    }

vector<ll> multiply(vector<ll> const& a, vector<ll> const& b) {
    vector<cd> fa(a.begin(), a.end()), fb(b.begin(), b.end());
    int n = 1;
    while(n < a.size() + b.size())
        n <<= 1;
    fa.resize(n);
    fb.resize(n);
    fft(fa);
    fft(fb);
    for(int i = 0; i < n; i++)
        fa[i] *= fb[i];
    fft(fa, -1);
    vector<ll> res(n);
    for(int i = 0; i < n; i++)
        res[i] = round(fa[i].real());
    return res;
}

```

6.9 Number Theoretic Transform

```

namespace FFT {
    /* ----- Adjust the constants here ----- */
    const int LN = 24; //23
    const int N = 1 << LN;
    typedef long long LL; // 2**23 * 119 + 1. 998244353
    // `MOD` must be of the form 2**`LN` * k + 1, where k is odd.
    const LL MOD = 9223372036737335297; // 2**24 * 54975513881 + 1.
    const LL PRIMITIVE_ROOT = 3; // Primitive root modulo `MOD`.
    /* ----- End of constants ----- */
}

```

```

LL root[N];
inline LL power(LL x, LL y) {
    LL ret = 1;
    for(; y >>= 1) {
        if(y & 1)
            ret = (__int128) ret * x % MOD;
        x = (__int128) x * x % MOD;
    }
    return ret;
}
inline void init_fft() {
    const LL UNITY = power(PRIMITIVE_ROOT, MOD - 1 >> LN);
    root[0] = 1;
    for(int i = 1; i < N; i++)
        root[i] = (__int128) UNITY * root[i - 1] % MOD;
    return;
}
// n = 2^k is the length of polynom
inline void fft(int n, vector<LL>& a, bool invert) {
    for(int i = 1, j = 0; i < n; i++) {
        int bit = n >> 1;
        for(; j >= bit; bit >>= 1)
            j -= bit;
        j += bit;
        if(i < j)
            swap(a[i], a[j]);
    }
    for(int len = 2; len <= n; len <<= 1) {
        LL wlen = (invert ? root[N - N / len] : root[N / len]);
        for(int i = 0; i < n; i += len) {
            LL w = 1;
            for(int j = 0; j < len >> 1; j++) {
                LL u = a[i + j];
                LL v = (__int128) a[i + j + len / 2] * w % MOD;
                a[i + j] = ((__int128) u + v) % MOD;
                a[i + j + len / 2] = ((__int128) u - v + MOD) % MOD;
                w = (__int128) w * wlen % MOD;
            }
        }
    }
    if(invert) {
        LL inv = power(n, MOD - 2);
        for(int i = 0; i < n; i++)
            a[i] = (__int128) a[i] * inv % MOD;
    }
    return;
}
inline vector<LL> multiply(vector<LL> a, vector<LL> b) {
    vector<LL> c;
    int len = 1 << 32 - __builtin_clz(a.size() + b.size() - 1);
    a.resize(len, 0);
    b.resize(len, 0);
    fft(len, a, false);
    fft(len, b, false);
    c.resize(len);
    for(int i = 0; i < len; i++)
        c[i] = (__int128) a[i] * b[i] % MOD;
    fft(len, c, true);
    return c;
}
//FFT::init_fft(); wajib di panggil init di awal
}

```

6.10 Gauss-Jordan

```
// Gauss-Jordan elimination with full pivoting.
//
// Uses:
// (1) solving systems of linear equations (AX=B)
// (2) inverting matrices (AX=I)
// (3) computing determinants of square matrices
//
// Running time: O(n^3)
//
// INPUT:  a[][] = an nxn matrix
//         b[][] = an nxm matrix
//
// OUTPUT:  X      = an nxm matrix (stored in b[][])
//         A^[-1] = an nxn matrix (stored in a[][])
//         returns determinant of a[][]
const double EPS = 1e-10;

typedef vector<int> VI;
typedef double T;
typedef vector<T> VT;
typedef vector<VT> VVT;
T GaussJordan(VVT& a, VVT& b) {
    const int n = a.size();
    const int m = b[0].size();
    VI irow(n), icol(n), ipiv(n);
    T det = 1;
    for(int i = 0; i < n; i++) {
        int pj = -1, pk = -1;
        for(int j = 0; j < n; j++) if(!ipiv[j])
            for(int k = 0; k < n; k++) if(!ipiv[k])
                if(pj == -1 || fabs(a[pj][k]) > fabs(a[pj][pk])) {
                    pk = k;
                    pj = j;
                    pk = k;
                }
        if(fabs(a[pj][pk]) < EPS) {
            cerr << "Matrix is singular." << endl;
            exit(0);
        }
        ipiv[pj]++;
        swap(a[pj], a[pk]);
        swap(b[pj], b[pk]);
        if(pj != pk)
            det *= -1;
        irow[i] = pj;
        icol[i] = pk;
        T c = 1.0 / a[pk][pk];
        det *= a[pk][pk];
        a[pk][pk] = 1.0;
        for(int p = 0; p < n; p++)
            a[pk][p] *= c;
        for(int p = 0; p < m; p++)
            b[pk][p] *= c;
        for(int p = 0; p < n; p++) if(p != pk) {
            c = a[p][pk];
            a[p][pk] = 0;
            for(int q = 0; q < n; q++)
                a[p][q] -= a[pk][q] * c;
            for(int q = 0; q < m; q++)
                b[p][q] -= b[pk][q] * c;
        }
    }
    for(int p = n - 1; p >= 0; p--) if(irow[p] != icol[p]) {
        for(int k = 0; k < n; k++)
            swap(a[k][irow[p]], a[k][icol[p]]);
    }
    return det;
}
```

```
int main() {
    const int n = 4;
    const int m = 2;
    double A[n][n] = { {1, 2, 3, 4}, {1, 0, 1, 0}, {5, 3, 2, 4}, {6, 1, 4, 6} };
    double B[n][m] = { {1, 2}, {4, 3}, {5, 6}, {8, 7} };
    VVT a(n), b(n);
    for(int i = 0; i < n; i++) {
        a[i] = VT(A[i], A[i] + n);
        b[i] = VT(B[i], B[i] + m);
    }
    double det = GaussJordan(a, b);
    // expected: 60
    cout << "Determinant: " << det << endl;
    // expected: -0.233333 0.166667 0.133333 0.066667
    //              0.166667 0.166667 0.333333 -0.333333
    //              0.233333 0.833333 -0.133333 -0.066667
    //              0.05 -0.75 -0.1 0.2
    cout << "Inverse: " << endl;
    for(int i = 0; i < n; i++) {
        for(int j = 0; j < n; j++)
            cout << a[i][j] << ' ';
        cout << endl;
    }
    // expected: 1.63333 1.3
    //              -0.166667 0.5
    //              2.36667 1.7
    //              -1.85 -1.35
    cout << "Solution: " << endl;
    for(int i = 0; i < n; i++) {
        for(int j = 0; j < m; j++)
            cout << b[i][j] << ' ';
        cout << endl;
    }
}
```

6.11 Derangement

```
der[0] = 1;
der[1] = 0;
for(int i = 2; i <= 10; ++i)
    der[i] = (i-1) * (der[i-1] + der[i-2]);
```

6.12 Bernoulli Number

$$\sum_{k=1}^n k^m = \frac{1}{m+1} \sum_{i=0}^m \binom{m+1}{i} B_i^+ n^{m+1-i} = m! \sum_{i=0}^m \frac{B_i^+ n^{m+1-i}}{i!(m+1-i)!}$$

$$B_n^+ = 1 - \sum_{i=0}^{n-1} \binom{n}{i} \frac{B_i^+}{n-i+1}, \quad B_0^+ = 1$$

6.13 Forbenius Number

(X* Y) - (X + Y) and total count is (X - 1) * (Y - 1) / 2

6.14 Stars and Bars with Upper Bound

$$P = (1 - X^{r_1+1}) \dots (1 - X^{r_n+1}) = \sum_i c_i X^{e_i}$$

$$Ans = \sum_i c_i \binom{N-e_i+n-1}{n-1}$$

6.15 Arithmetic Sequences

$$U_n = a + (n-1)a_1 + \frac{(n-1)(n-2)}{1 \times 2} a_2 + \dots + \frac{(n-1)(n-2)(n-3) \dots}{1 \times 2 \times 3 \dots} a_r$$

$$S_n = n \times a + \frac{n(n-1)}{1 \times 2} a_1 + \frac{n(n-1)(n-2)}{1 \times 2 \times 3} a_2 + \dots + \frac{n(n-1)(n-2)(n-3) \dots}{1 \times 2 \times 3 \dots} a_r$$

6.16 FWHT

```
// Desc : Transform a polynom to obtain a_i * b_j * x^
//         ^ (i XOR j) or combinations
// Time : O(N log N) with N = 2^K
// OP => c00 c01 c10 c11 | c00 c01 c10 c11 inv
// XOR => +1 +1 +1 -1 | +1 +1 +1 -1 | div the ←
//         inverse with size = n
// AND => 1 +1 0 1 | 1 -1 0 1 | no ←
//         comment
// OR => 1 0 +1 1 | 1 0 -1 1 | no ←
//         comment
typedef vector<long long> vec;
void FWHT(vec& a) {
    int n = a.size();
    for(int lvl = 1; 2 * lvl <= n; lvl <= 1) {
        for(int i = 0; i < n; i += 2 * lvl) {
            for(int j = 0; j < lvl; j++) { // do not forget ←
                to modulo
                long long u = a[i + j], v = a[i + lvl + j];
                a[i + j] = u + v; // c00 * u + c01 * v
                a[i + lvl + j] = u - v; // c10 * u + c11 * v
            }
        }
    }
} // you can convolve as usual
```

6.17 Basis Vector

```
int basis[d]; // basis[i] keeps the mask of the vector ←
// whose f value is i
int sz; // Current size of the basis
void insertVector(int mask) {
    for(int i = 0; i < d; i++) {
        if((mask & 1 << i) == 0) {
            continue; // continue if i != f(mask)
        }
        if(!basis[i]) { // If there is no basis vector ←
            with the i'th bit set,
            // then insert this vector into the basis
            basis[i] = mask;
            ++sz;
            return;
        }
        mask ^= basis[i]; // Otherwise subtract the basis ←
        // vector from this
    }
}
```